DAY 18 · THEORY HANDOUT

Run a Language Model With Ollama

Install the runner, download a model, and send a first local request.

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The idea and its useful terms

Install the runner, download a model, and send a first local request.

Ollama

A program that runs and manages local models.

Model tag

A name and version label such as llama3.2:latest.

Local server

A program waiting for requests on this computer.

The model file contains learned numbers. Ollama loads those numbers and runs the model; it is not the model itself.

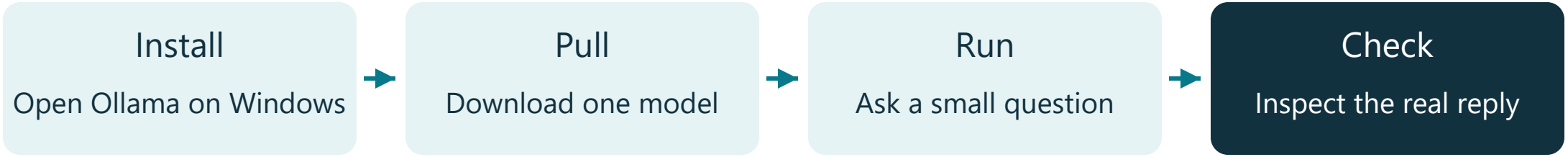
Local means generation happens on our computer in this lesson. Downloads need internet. Optional cloud or web tools can still send data outside.

> If the server is stopped or the tag is missing, the request fails before the model can answer.

Follow the work, one step at a time

Each step has a job. We can inspect its output before moving on.

Read the path from left to right.



Each arrow means: pass the result to the next step.

A local server waits for a request at a host address and port. localhost is this computer; 11434 is Ollama's usual port. The model tag tells that server which downloaded model to load.

The chat request has a model name and a messages list. Each message has a role and content. The server returns reply text and run details. A REST API is this kind of request-and-response interface.

> If the answer is wrong, inspect the input and the intermediate results. The last sentence alone cannot show which step failed.

A small example we can check

The values here are classroom examples. The notebook shows actual results.

ollama list

See downloaded models.

No new model download.

ollama pull

Download a named model.

Size depends on its tag.

ollama run

Load and chat with it.

A first reply may be slow.

Read the given input first. Point to the fact or value needed for the requested output.

Run the matching example, then try the fresh input at the end of the notebook. Compare both results with the known input.

> If the server is stopped or the tag is missing, the request fails before the model can answer.

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Choose the setting or tool for its job

A different choice changes the work. These are the choices we compare.

Chat model

Generates a reply.

Use it for our first chat.

Embedding model

Returns number vectors.

Use it later for search.

Hardware

Bigger files need more RAM.

Choose a model that fits.

Check installed models before downloading another copy.

Ask one short question, then change the question and test again.

> Check the local chat endpoint in Python. localhost:11434 points to this computer, not a public website.

Practice, check, and read the full lesson

Use the notebook to try the steps. Use the article when you need the longer explanation.

1. Build

Run the small example.

Inspect each step.

2. Change

Try the fresh input.

Keep the goal clear.

3. Check

Compare with the source.

Keep mistakes visible.

Check the local chat endpoint in Python. localhost:11434 points to this computer, not a public website.

Read the complete tutorial breakdown for this section in our KGP Talkie article.

> If the server is stopped or the tag is missing, the request fails before the model can answer.

DAY 18 · COMPLETE

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